

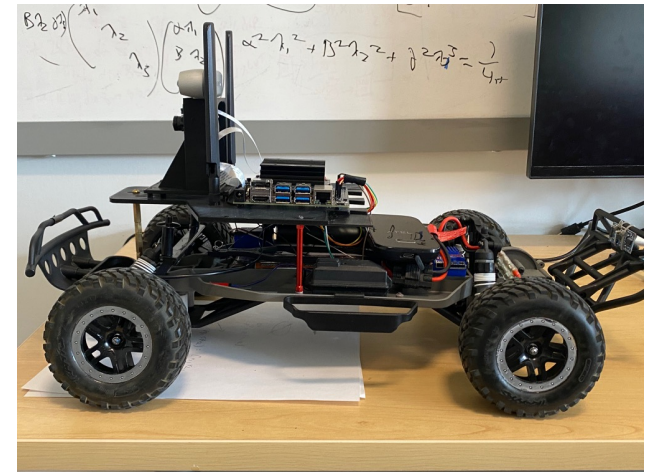
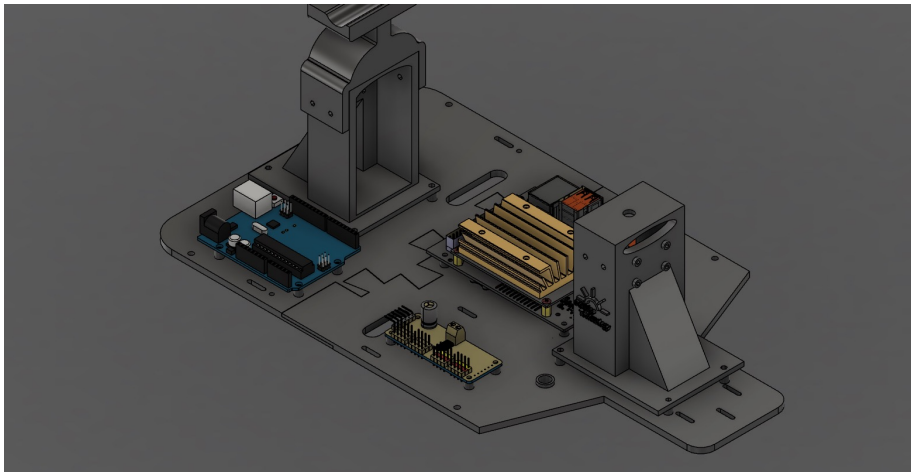
A Deep Neural Network-Based Autonomous System Emulation Toolkits for Self-Driving Vehicle

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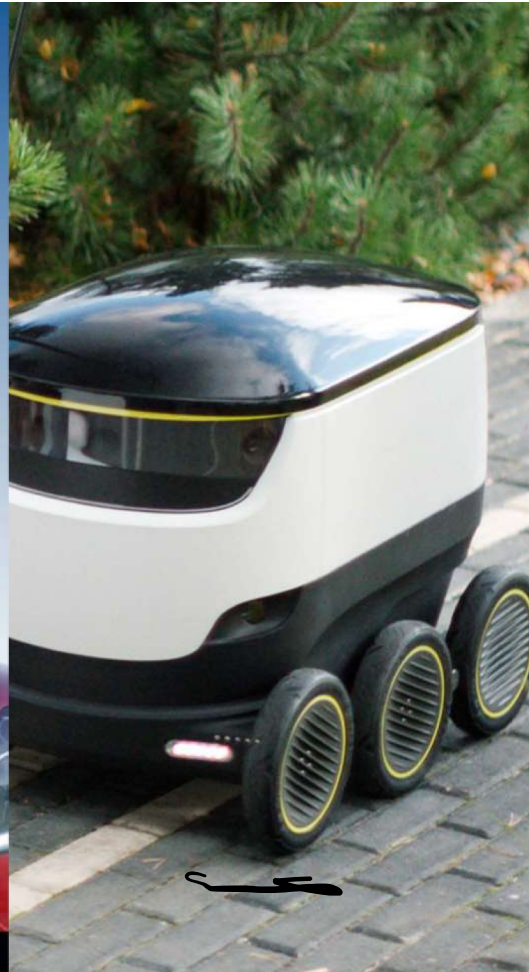




Cars



Trucks



Carts



Drones



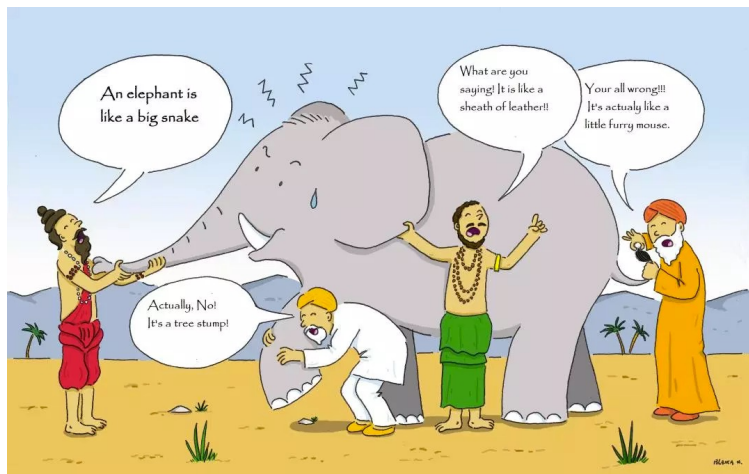
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- Self-Driving Introduction and Motivation
- Self-Driving in real vehicles and RC-car
- Self-Driving Perceptron (Brain)
 - Behavior Clone Based Algorithm
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- Demo 1: Build Reliable Self-driving Systems
- Demo 2: Obstacle Avoidance
- Self-Driving Challenges
- The drawback of Supervised Learning Algorithm
- Look into the Future and Conclusions

What is Self-Driving

Public Perception of V in Semi-Autonomo

- ❖ What is the Self-Driving?
- ❖ An autonomous vehicle or driverless vehicle can operate itself and perform necessary functions based on road conditions without any human intervention through ability to sense its surroundings.



SAE level	Name	Narrative Definition	Execution of Steering and Acceleration/Deceleration	Monitoring of Driving Environment	Fallback Performance of Dynamic Driving Task	System Capability (Driving Modes)
Human driver monitors the driving environment						
0	No Automation	the full-time performance by the human driver of all aspects of the dynamic driving task, even when enhanced by warning or intervention systems	Human driver	Human driver	Human driver	n/a
1	Driver Assistance	the driving mode-specific execution by a driver assistance system of either steering or acceleration/deceleration using information about the driving environment and with the expectation that the human driver perform all remaining aspects of the dynamic driving task	Human driver and system	Human driver	Human driver	Some driving modes
2	Partial Automation	the driving mode-specific execution by one or more driver assistance systems of both steering and acceleration/deceleration using information about the driving environment and with the expectation that the human driver perform all remaining aspects of the dynamic driving task	System	Human driver	Human driver	Some driving modes
Automated driving system ("system") monitors the driving environment						
3	Conditional Automation	the driving mode-specific performance by an automated driving system of all aspects of the dynamic driving task with the expectation that the human driver will respond appropriately to a request to intervene	System	System	Human driver	Some driving modes
4	High Automation	the driving mode-specific performance by an automated driving system of all aspects of the dynamic driving task, even if a human driver does not respond appropriately to a request to intervene	System	System	System	Some driving modes
5	Full Automation	the full-time performance by an automated driving system of all aspects of the dynamic driving task under all roadway and environmental conditions that can be managed by a human driver	System	System	System	All driving modes



AI → Deep Learning

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) AUTOMATION LEVELS

Full Automation



0

**No
Automation**

Zero autonomy; the driver performs all driving tasks.



1

**Driver
Assistance**

Vehicle is controlled by the driver, but some driving assist features may be included in the vehicle design.



2

**Partial
Automation**

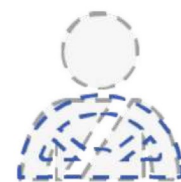
Vehicle has combined automated functions, like acceleration and steering, but the driver must remain engaged with the driving task and monitor the environment at all times.



3

**Conditional
Automation**

Driver is a necessity, but is not required to monitor the environment. The driver must be ready to take control of the vehicle at all times with notice.



4

**High
Automation**

The vehicle is capable of performing all driving functions under certain conditions. The driver may have the option to control the vehicle.



5

**Full
Automation**

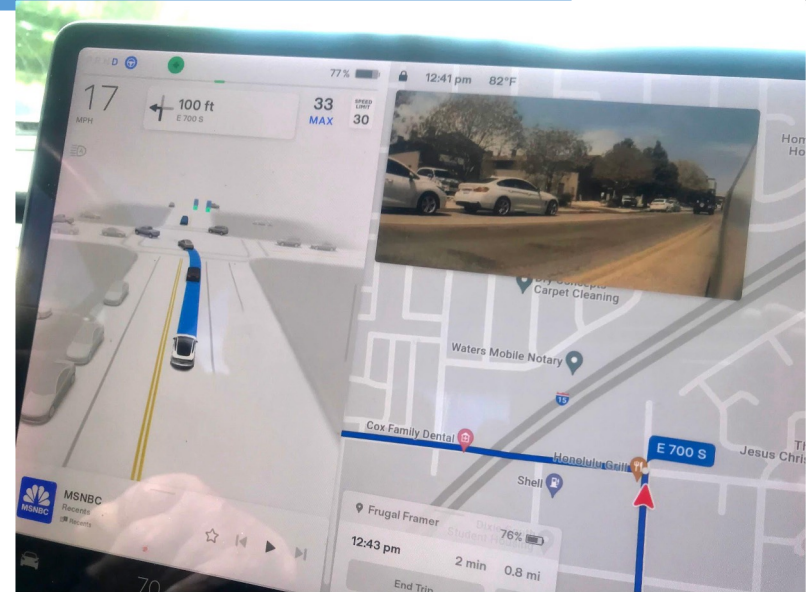
The vehicle is capable of performing all driving functions under all conditions. The driver may have the option to control the vehicle.

Some Facts

- 1.35 million deaths worldwide due to vehicle crashes
- 94% of crashes involve human choice or errors.
- Three million Americans age 40 and older are not able to drive due to their vision/blindness
- 79% of seniors age 65 and older living in car-dependent communities
- 42 hours in traffic jams each year per person

Machine Learning-based Self-Driving is a promising future of the workforce

- Autonomous Vehicle will Revolutionize our economy.
- The benefits of autonomous vehicles (AV) extend beyond safety – they could also greatly impact the labor market.
- AVs will create 300K US jobs in development, testing, maintenance, and deployment, as well as a need for technicians and mechanics with expertise in automation.
- <https://ts2.space/en/the-impact-of-autonomous-vehicles-on-employment-and-job-market/>



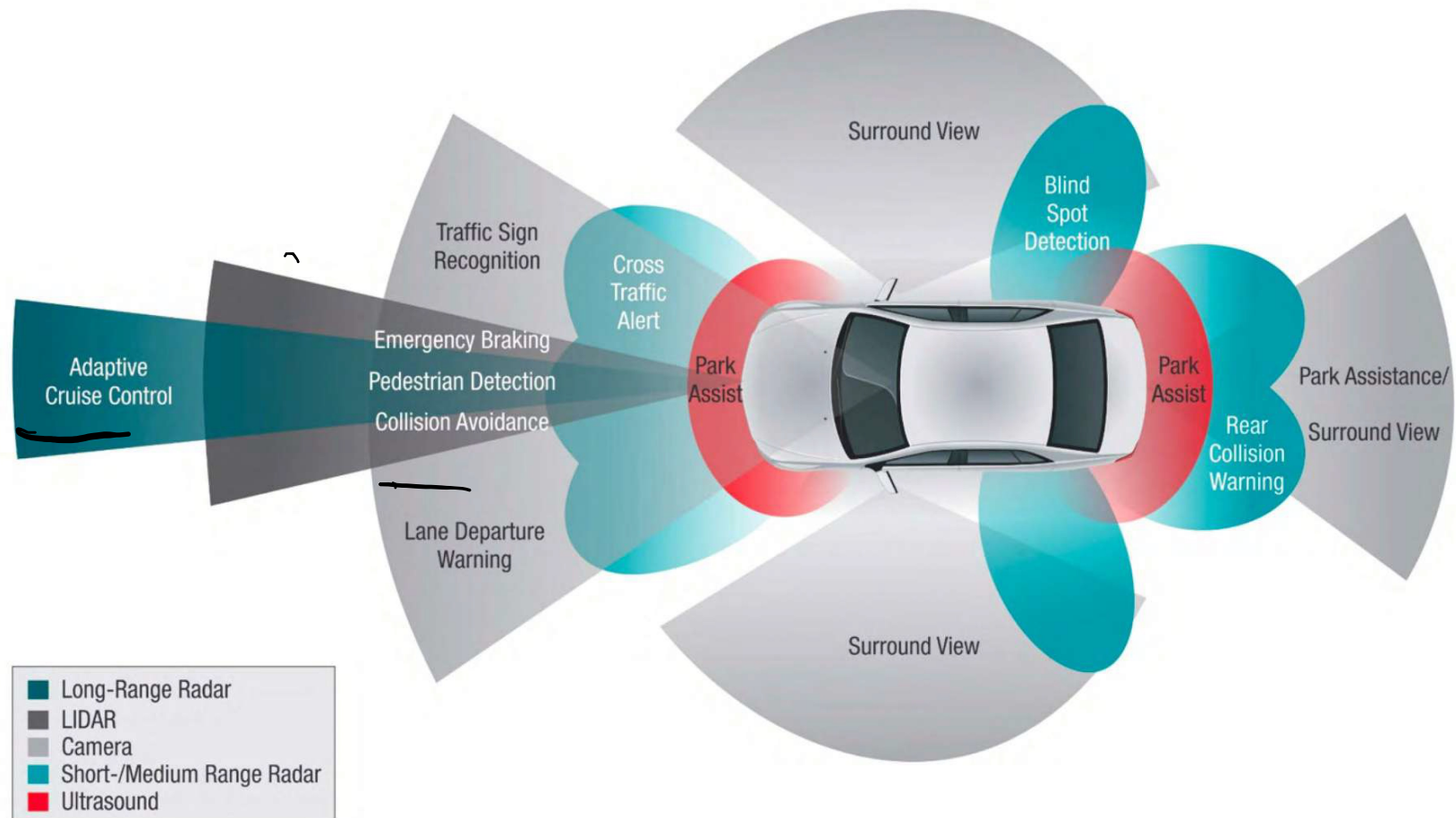
Economy Fraction of Autonomous Driving

- ❖ How big the industry will be (Look at the celebrity company Tesla)
- ❖ It is one of the most exciting/critical applications of AI/Machine learning
- ❖ Solving it will bring changes to society, climate change, less dependence on fossil fuel, and bringing the sharing economy to a new level.

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Tesla/WAYMO



Similar Dynamics, different scales

TRAXXAS XO-1



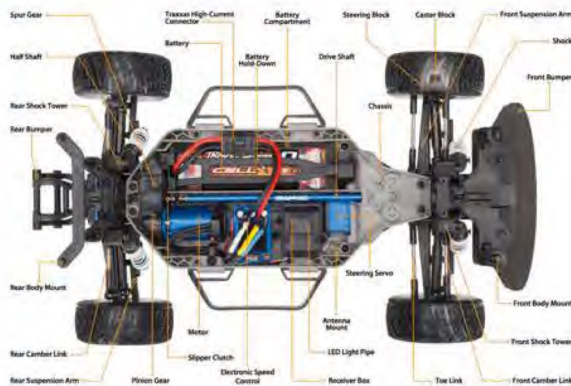
Tesla Model S



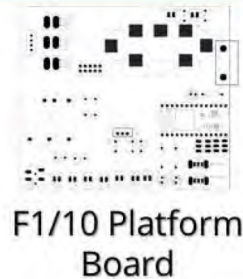
Traxxas 2WD Chassis



F1/10 Used in College Self-Driving Course (\$5000)



Chassis Design

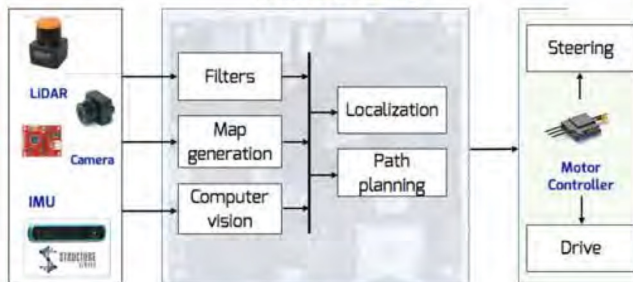


GPGPU Compute Platform
NVIDIA Jetson TX1/TX2

System Integration

Software Architecture

ROS



Perception Planning Control

GPU Accelerated Libraries



Research Enabled

Safe
Autonomy

Secure
Autonomy

Coordinated
Autonomy

Efficient
Autonomy

Lane Keeping Assist



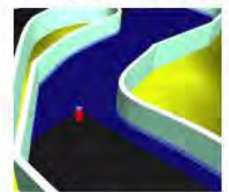
Model-Predictive Control

AV Data Collection



Simultaneous Localization And Mapping (SLAM)

F1/10 Simulator



DNN Racing

We designed Two Systems for the Same Goal.

- We use two different hardware systems to implement autonomous driving.
- Enables flexibility in design and lowers the learning barriers

Jetson

- Powerful embedded GPU
- ROS support for messaging
- Depth Camera

Smart Phone

- Application based, very cheap
- NPU
- High camera resolution

iPhone → Support LiDAR and 3-D scanning

Openbot Based Self-Driving Car

Sensor Integration



LiDAR



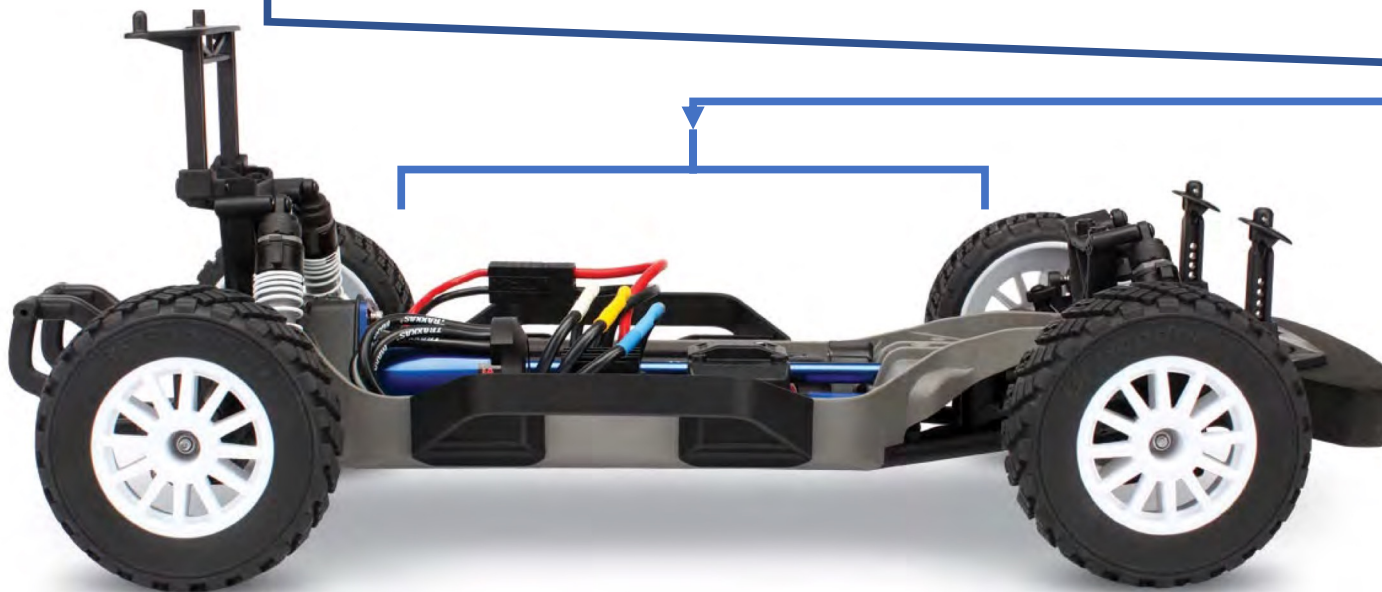
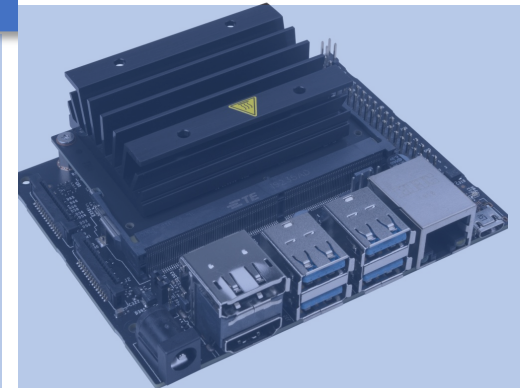
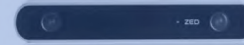
Camera



IMU



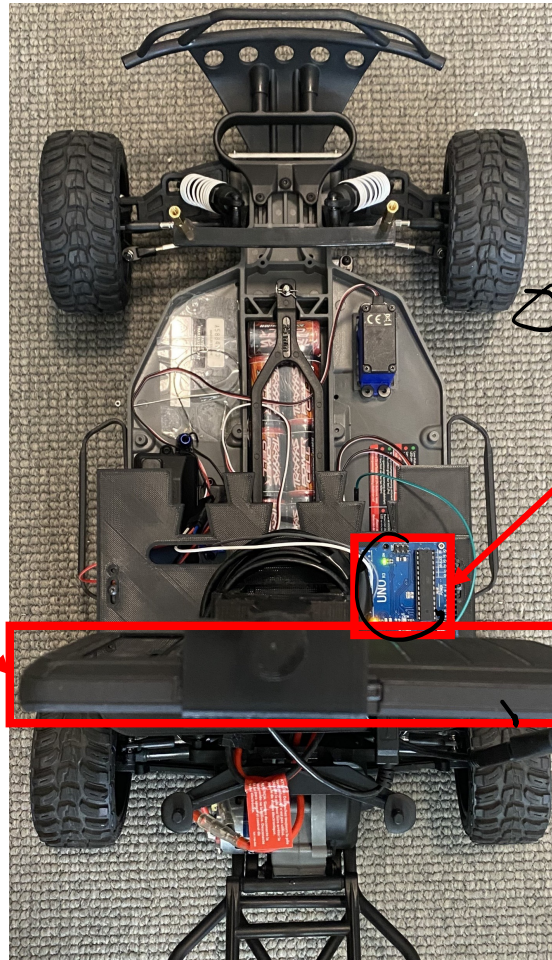
IR Depth Cameras



Open Bot Hardware \$500

Samsung S23/iPhone

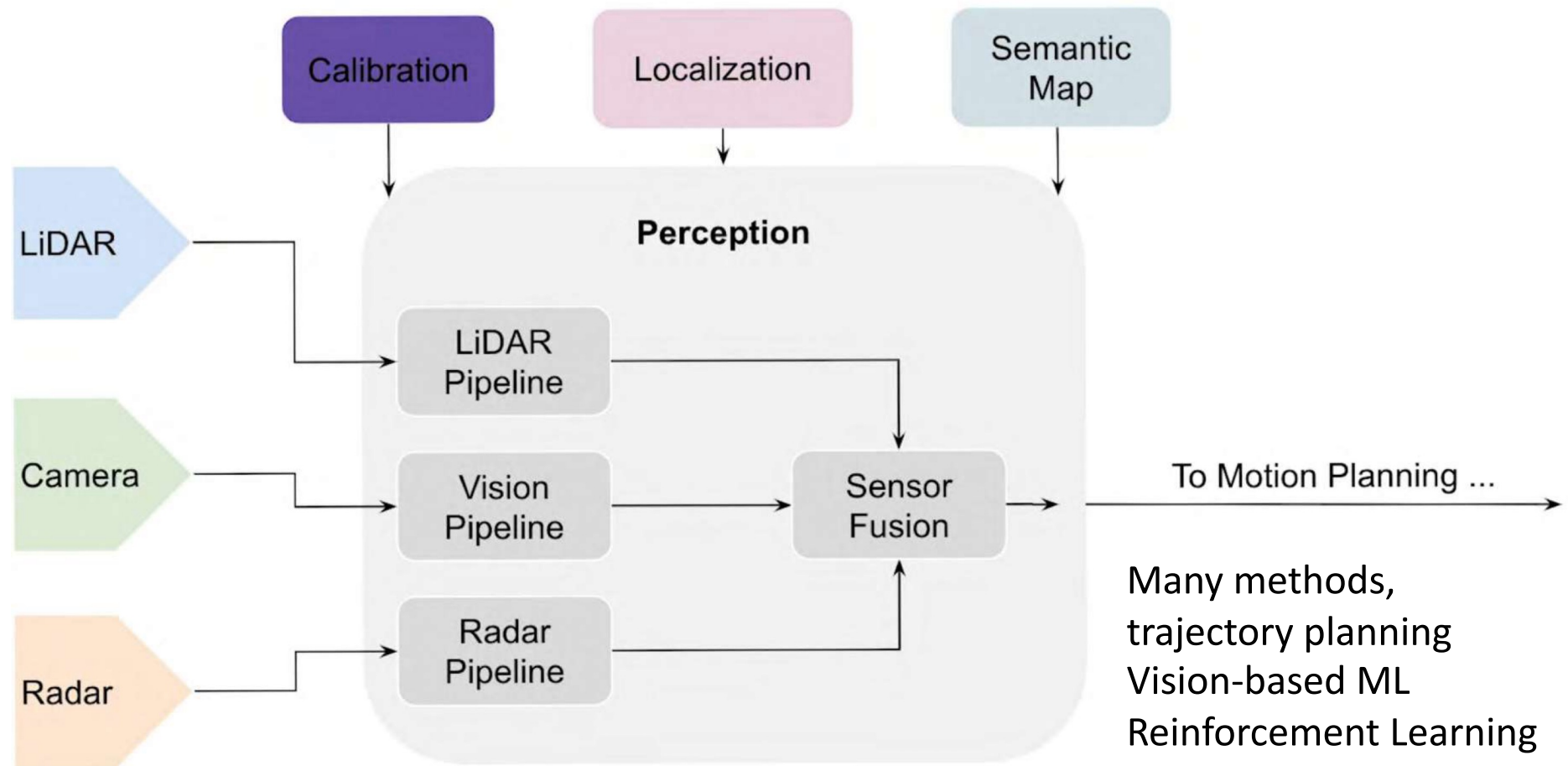
Arduino UNO



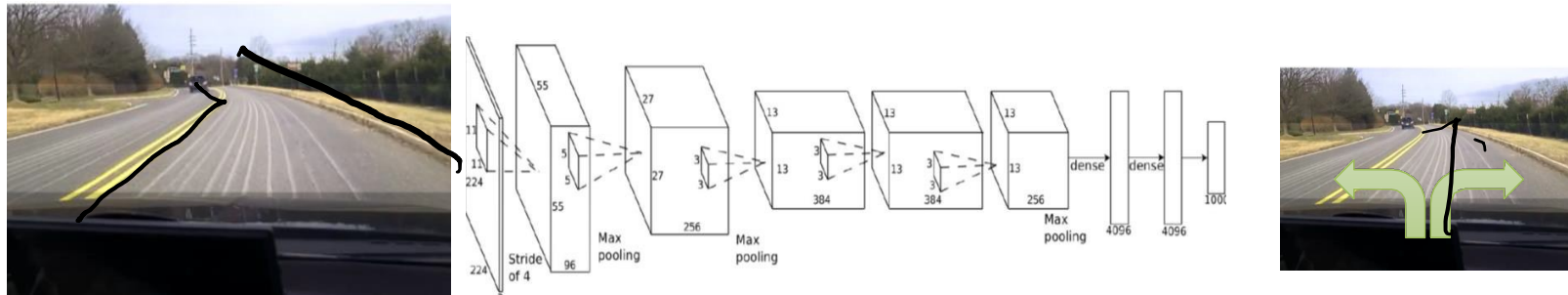
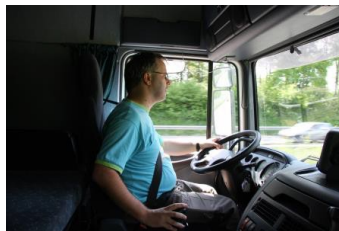
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Perceptions in AV Stack



Imitation Learning


 $\mathbf{o}_t$
 $\pi_{\theta}(\mathbf{a}_t | \mathbf{o}_t)$
 $\mathbf{a}_t$


Images: Bojarski et al. '16, NVIDIA

 $\mathbf{o}_t, \mathbf{a}_t$

training data

supervised learning

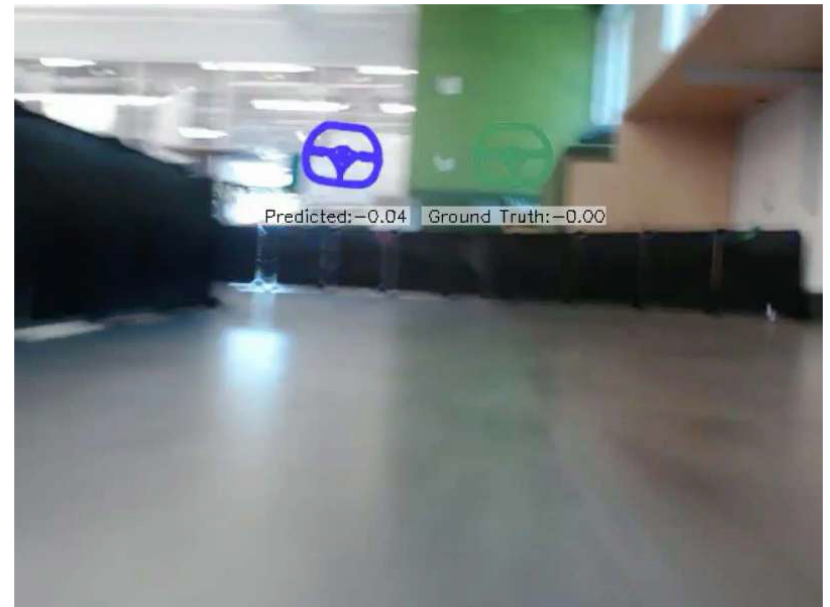
 $\pi_{\theta}(\mathbf{a}_t | \mathbf{o}_t)$

End-to-End Driving

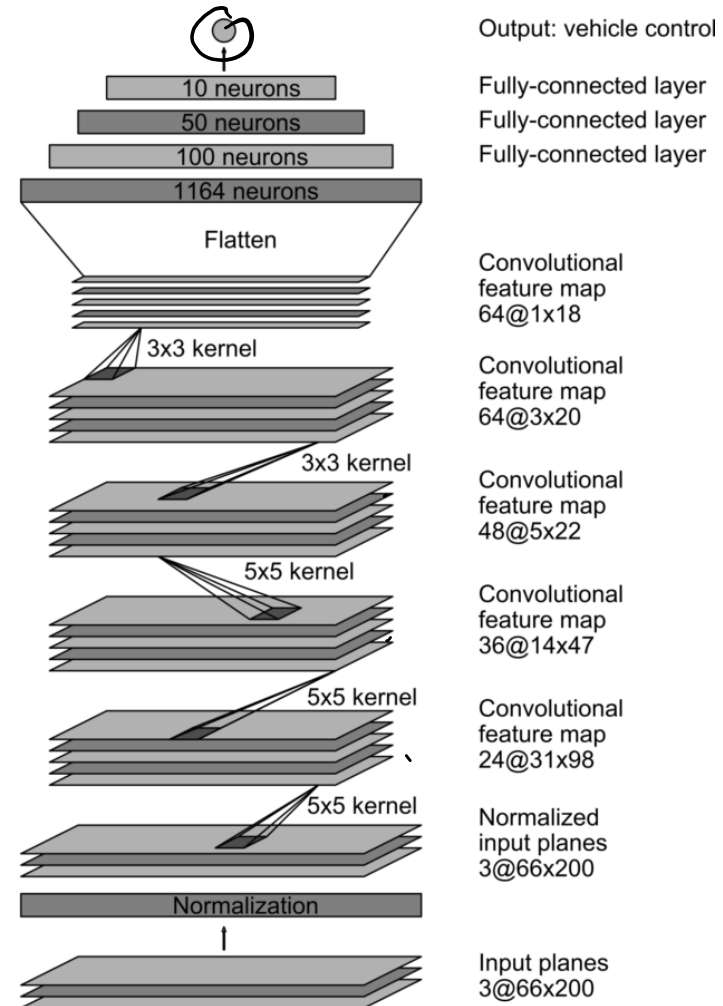
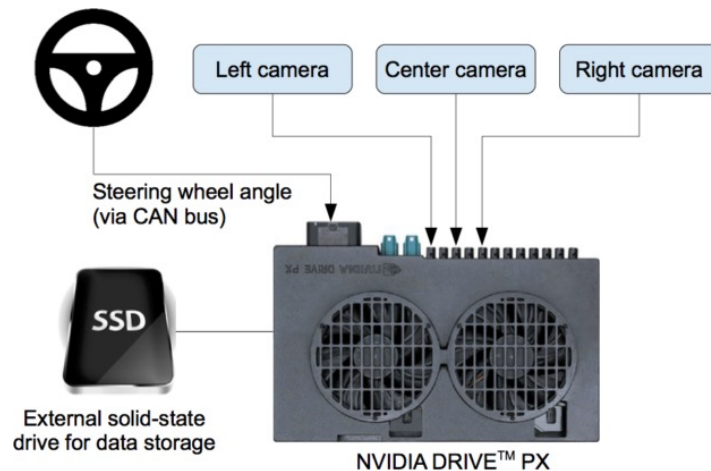
Problem:

End-to-End driving with imitation learning and Behavior Clones.

1. Train a network to map pixels to steering angle
2. Pixels to trajectory and throttles

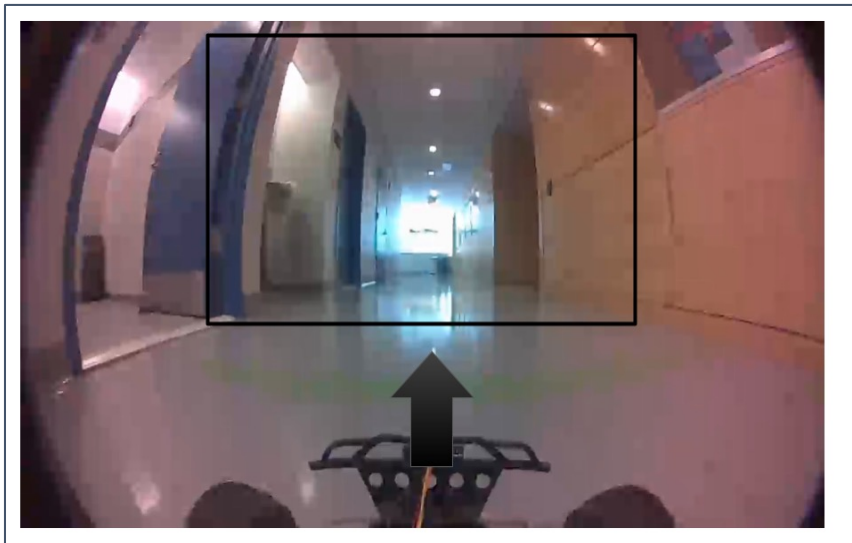


Behavior Cloning with Nvidia PilotNet

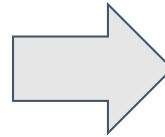


Behavior Cloning : Data Augmentation

An example is shown below, the black box is the clipped image to be input to the model and the arrow is the adjusted steering angle.



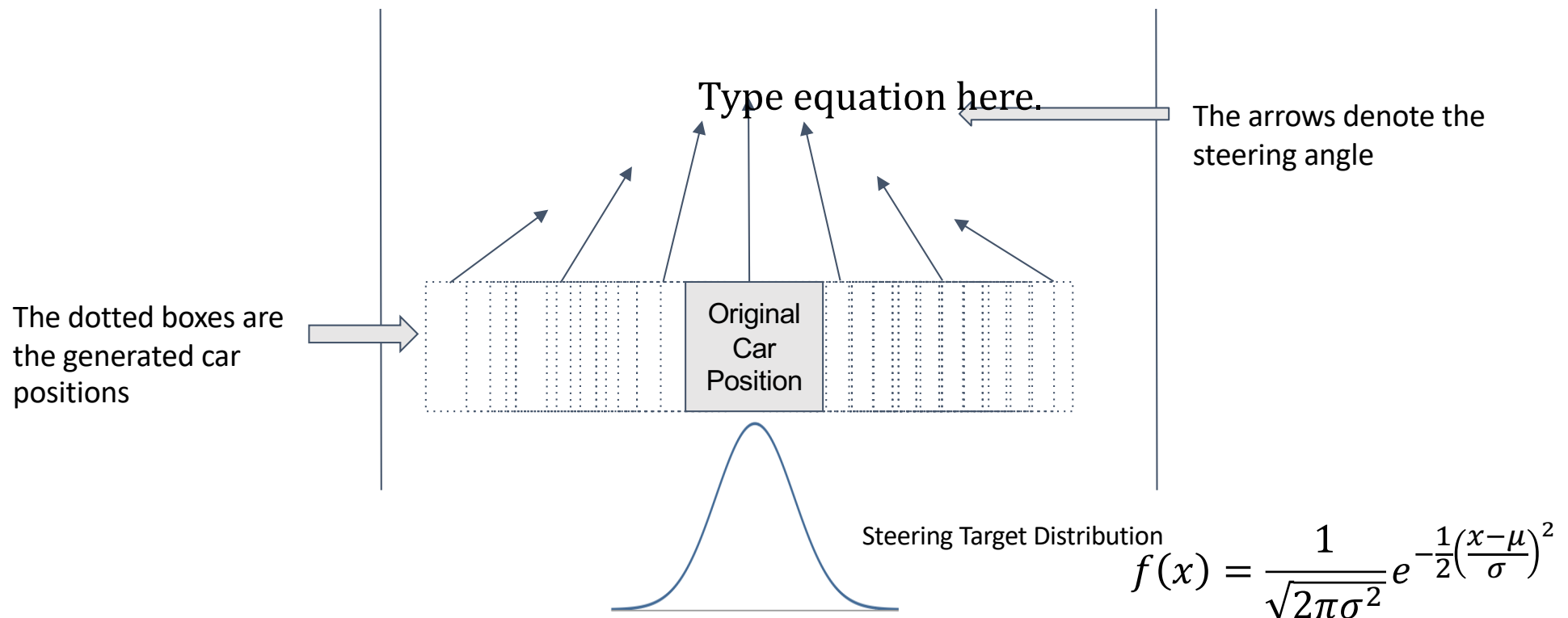
Steering angle of 0.0



Steering angle of -0.5

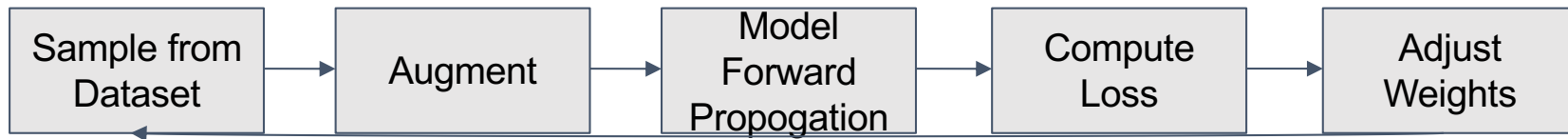
Behavior Cloning : Data Augmentation

In this way we can simulate the left/right orientations of the car at every data point.



Behavior Cloning : Training

Training the neural network is fairly straightforward; we randomly sample from the dataset, augment the data, then input it into the model.



The model produces steering and throttle predictions that we can test against the actual steering and throttle we recorded (or generated).

Using this, we compute the loss, and through backpropagation, we adjust the weights.

Behavior Cloning : Loss

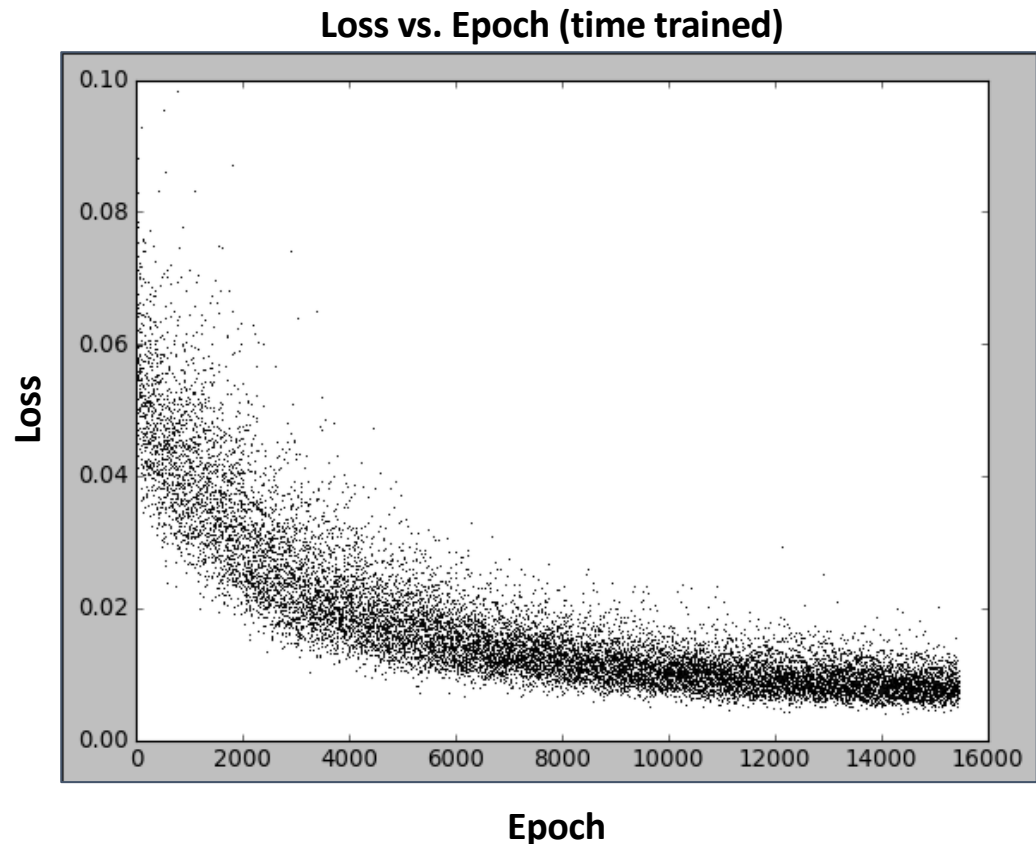
Loss is a metric by which we can test the validity of the model, there are many many functions that can be used. For our purposes we use MSE Loss (Mean Squared Error):

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2$$

This function simply takes the error at each point:

$$y_{\text{actual}} - y_{\text{predicted}}$$

Then squares and averages over all of them. This penalizes the model more for larger errors than for smaller ones. The graph shown is the loss as the model trains. As the loss tends towards 0 the model gets better and better at the task:

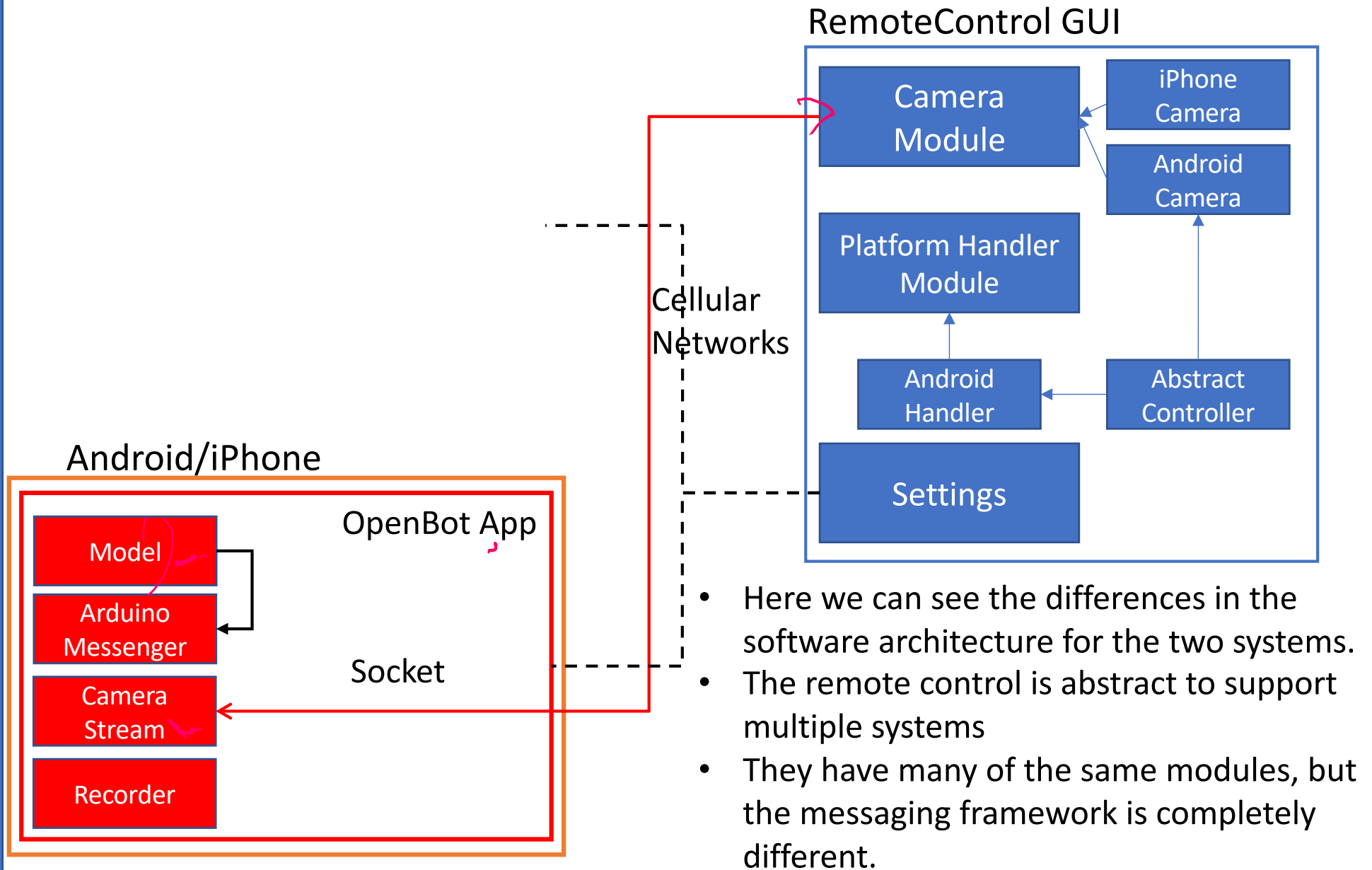


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Demo of Openbot self-driving framework





- Here we can see the differences in the software architecture for the two systems.
- The remote control is abstract to support multiple systems
- They have many of the same modules, but the messaging framework is completely different.

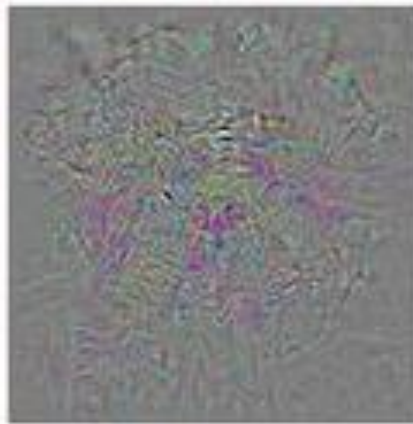
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Challenge: Self-Driving Security



Prediction: **Dog**



+ Distortion

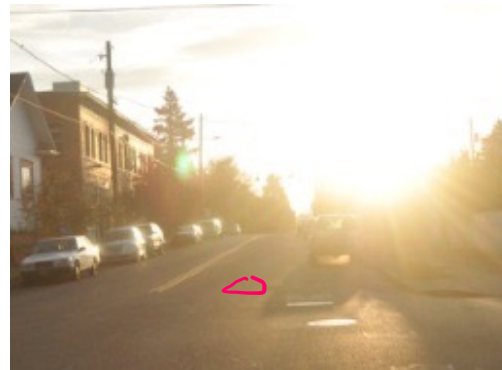


Prediction: **Ostrich**



Prediction: **Semi-Truck**

+



+ Attacks



Prediction: **Clouds**

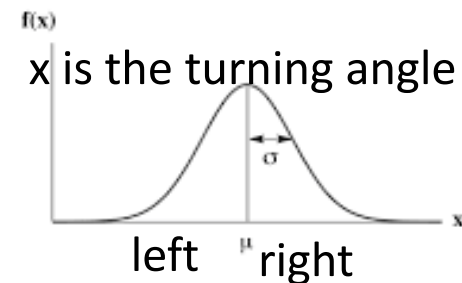
Challenge: Will Behavior Clone Work?

Small Mismatch will accumulate

Corrective actions

Discrete Action is OK, Soft-max on Left or Right

What about Continuous Actions?



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

$e = 2.71828$



Look into Future

- The question: When will full autonomous driving happen?
- What are the potential intermediate steps:
 - A) Algorithm Development for an Explainable AI system (phantom brake in Tesla)
 - B) Education and Outreach for public acceptance, Crowd-source, and Crowd development)
 - C) Infrastructure support: Star-Link and 5G/6G (99.999% wireless networks) to allow remote monitoring and control
 - D) Basic IT infrastructure development (fast, low-latency communications).

Reinforcement learning provides a formalism for *behavior*

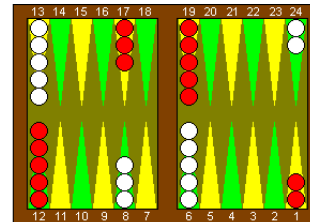
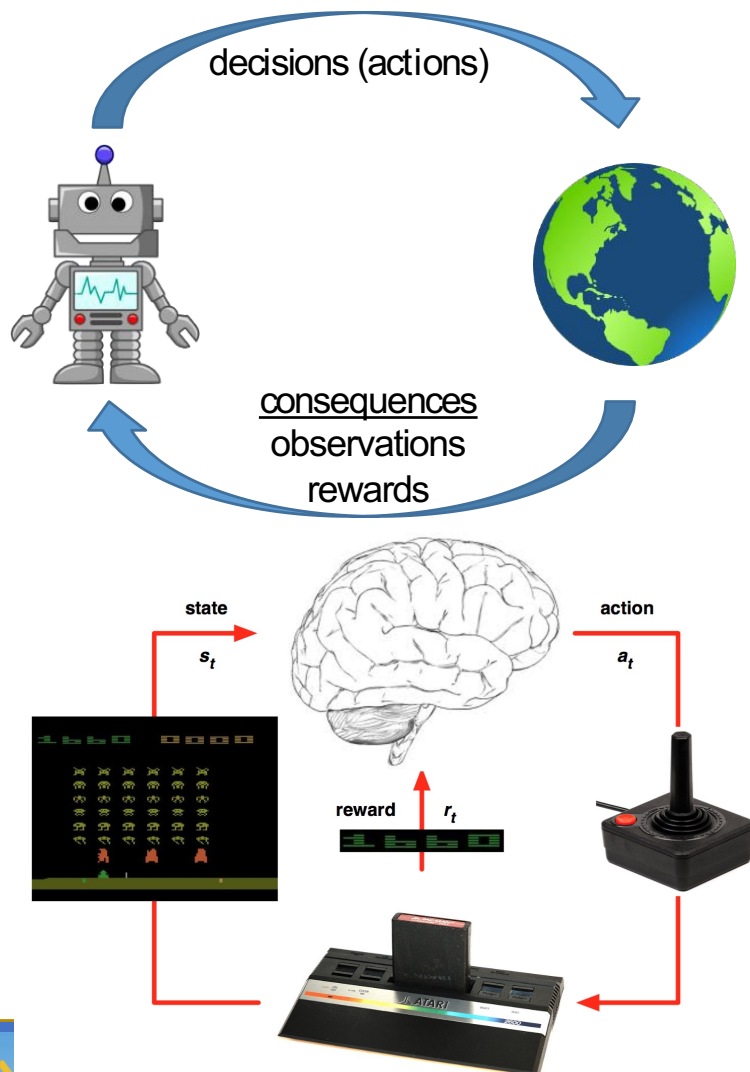
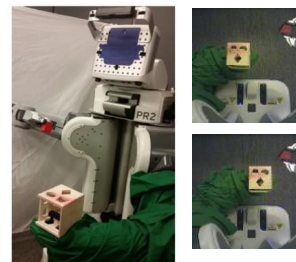
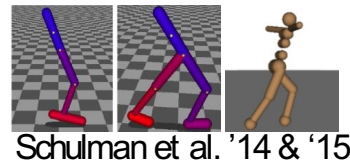
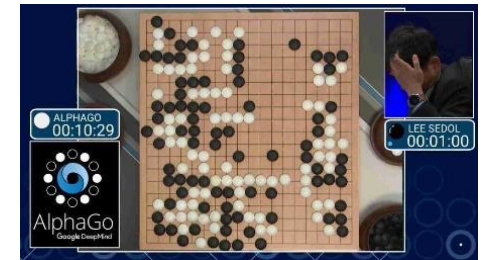


Figure 2. An illustration of the normal opening position in backgammon. TD-Gammon has sparked a near-universal conversion in the way experts play certain opening rolls. For example, with an opening roll of 4-1, most players have now switched from the traditional move of 13-9, 6-5, to TD-Gammon's preference, 13-9, 24-23. TD-Gammon's analysis is given in Table 2.

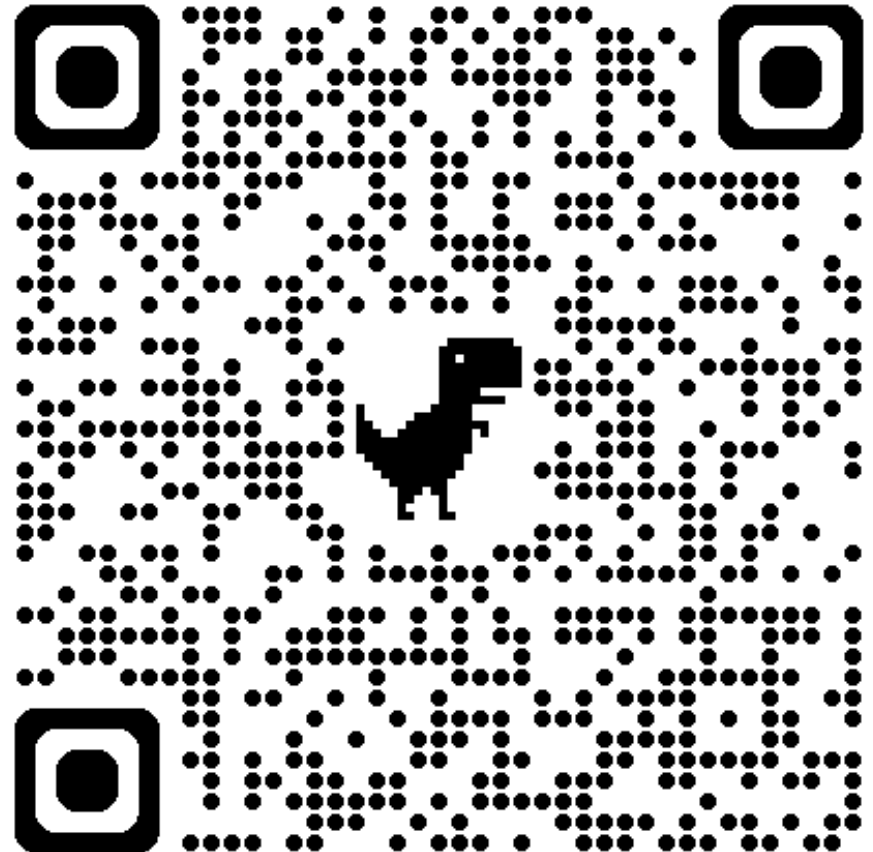


Conclusions

- Self-Driving depends on an accurate learning model. Small errors accumulate and lead to substantial deviations from the track.
- No sensor works 100%; multiple sensors add reliability.
- Representation (embedding) is the key to uncovering the state information of observations and enhancing learning capability.
- Large training data is needed to mitigate the limitation of a supervised learning model.
- Performance and QoS are critical in self-driving: low power consumption, real-time decision making, and the flexibility to support transfer learning: incremental training, and independence from cloud and data center.
- Planning/controlling autonomous vehicles involves decision makings continuously in an episode to maximize future values rather than clone experts' behavior.

<https://forms.gle/3EVJ1RMeMMe3mgENA>

Thank
You!

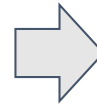


<https://colab.research.google.com/drive/1KxivPKIVEO6aLZMf56LEOsrIAHRInqRO>

Behavior Cloning : Data Augmentation

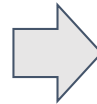
Flipping: Flip the image horizontally and flip the steering angle to match

Equally trains left and right, further reduces bias



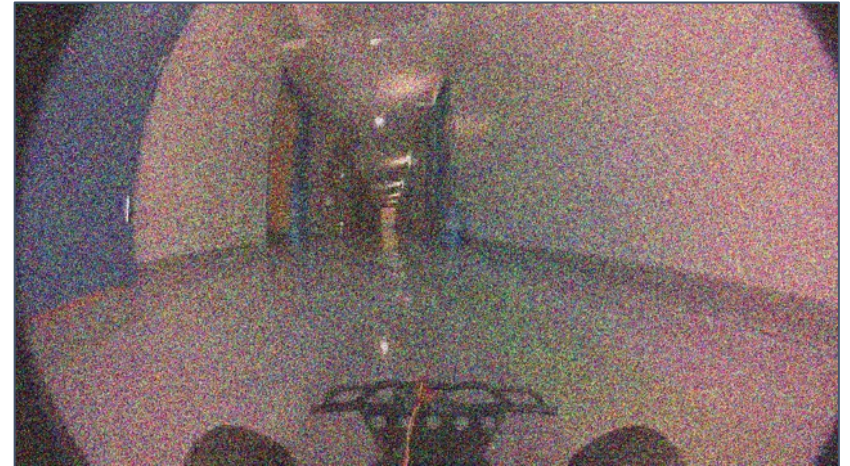
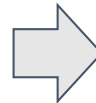
Behavior Cloning : Data Augmentation

Lighting - Randomly adjust the contrast and brightness
Used to increase robustness of the model to lighting effects



Behavior Cloning : Data Augmentation

Noise - add gaussian noise to the image



Behavior Cloning : Data Augmentation

Grayscale - although not used, the option to convert the rgb image to grayscale remains in our training notebook. Furthermore, you can also equalize the histograms



This can be useful for reducing the computation cost associated with the channels of the image or for reducing the model's dependency on recognizing certain colors.